

Problem set 2
Due Thursday Jan 29, 11:59pm

Exercise 2.1 (3 points). Let F be any ordered field. Suppose $A \subset F$ has a supremum. Suppose $\epsilon \in F$ with $\epsilon > 0$. Prove that there exists some $x \in A$ such that $x > (\sup A) - \epsilon$.

Exercise 2.2 (5 points; Rudin 1.5). Let $A \subset \mathbb{R}$ be nonempty and bounded below. Define

$$-A = \{-x \mid x \in A\}.$$

Prove that $\inf A = -\sup(-A)$.

Exercise 2.3 (10 points). For each of the following, A is a subset of $\mathbb{Q}$. Answer the following in each case, and prove your answers:

- Is A bounded above?
- Does A have a maximum element, and if so, what is it?
- Does A have a supremum in $\mathbb{Q}$, and if so, what is it?

(1) $A = \{-\frac{1}{n} \mid n \in \mathbb{N}\}.$

(2) $A = \{x \in \mathbb{Q} \mid 0 < x < 1\}.$

(3) $A = \{x \in \mathbb{Q} \mid 0 < x < t\}$ where t is a positive irrational number.

Exercise 2.4 (5 points). This exercise is a reminder that the cardinality of sets is sometimes a counterintuitive notion of size.

- (1) Suppose $a, b \in \mathbb{R}$, with $a < b$. Prove that there is a bijection from (a, b) to $(0, 1)$.
- (2) Prove that there is a bijection from $(0, 1)$ to $(1, \infty)$.
- (3) Suppose $a, b \in \mathbb{R}$. Prove that there is a bijection from (a, ∞) to (b, ∞) .

Exercise 2.5 (10 points). Decide if the following sets are countable or uncountable. Prove your answers.

- (1) The set of irrational numbers.
- (2) The set of sequences of natural numbers $n_1, n_2, n_3, \dots$
- (3) The set of sequences of natural numbers $n_1, n_2, n_3, \dots$ such that there exists some $l \in \mathbb{N}$ such that $n_k = 0$ for all $k > l$.

Exercise 2.6 (optional, not for credit; Rudin 1.8 and 1.9). Define the complex numbers as the set

$$\mathbb{C} = \{a + bi \mid a, b \in \mathbb{R}\}$$

with the addition and multiplication rules given by

- $(a + bi) + (a' + b'i) = (a + a') + (b + b')i,$
- $(a + bi)(a' + b'i) = (aa' - bb') + (ab' + a'b)i.$

- (1) Prove that $\mathbb{C}$ with these operations (and an appropriate 0, 1, negation, inversion) is a field.
- (2) Prove that $\mathbb{C}$ with these operations cannot be made into an ordered field.
- (3) Define the *lexicographic order* on $\mathbb{C}$ by defining

$$a + bi < c + di \iff a < c \text{ or } (a = c \text{ and } b < d).$$

Prove that this makes $\mathbb{C}$ into an ordered *set* (though not an ordered *field*).

(4) Does the lexicographic order on $\mathbb{C}$ have the least upper bound property?